Measuring the Reset Distribution: A Tolerance Radius, Referenced to Open-Loop Replay

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***Abstract*—Two choices beneath a robot policy are rarely reported: where an evaluation places the object at reset relative to the demonstrations, and how a fixed budget of demonstrations is distributed. On a low-cost arm with one imitation-learning method (ACT), we place every reset within 2 mm of a target (median error 0.3 mm), log it, and report success against the distance to the nearest demonstration, summarised by a *tolerance radius*: the distance at which success is halfway between its levels near and far from the demonstrations. An open-loop replay of demonstrations has a radius of 16.5 mm (95 % interval 13.0 to 21.5); resets placed by eye from a reference image miss by a median of 15.9 mm. The policy succeeds at larger distances than the replay, but its success depends on location as well as distance (28 of 36 trials inside the convex hull of its demonstrations, 9 of 24 outside), and its fitted radius of 39.0 mm is not established under a pre-specified robustness check. A policy trained on 55 distinct start positions succeeded at 22 of 26 shared test placements, and one trained on 5 demonstrations at each of 11 locations at 16; the paired test does not establish a difference ($p = 0.11$), and the spread layout's own radius was not measured. We propose five reporting-checklist entries.**

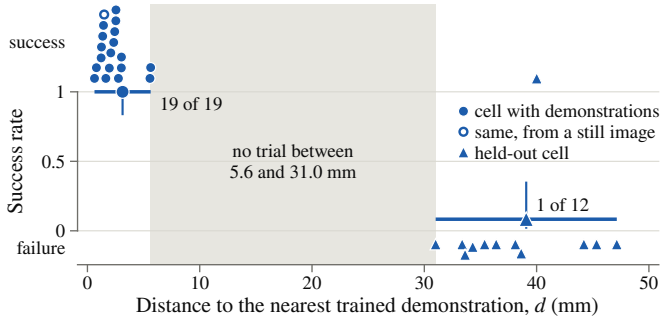


Fig. 1. Our earlier evaluation of the same task, re-analysed. Each mark is one of 31 trials (12 scenes, each run two or three times): its outcome against the distance from the cube's start position to the nearest demonstration the policy was trained on. Large marks: each group's success rate, with a horizontal line over its range of distances and a vertical line for its Wilson 95 % interval. A held-out cell is a grid cell without demonstrations. Start positions were not logged; we measured them afterwards from the trial recordings (open circle: the one trial measured from a still image, not from its video). No trial started between 5.6 and 31 mm.

I. Introduction

Two choices beneath a robot policy are rarely reported: where an evaluation places the object at reset relative to the demonstrations, and how a fixed budget of demonstrations is distributed. We study both through success as a function of the distance to the nearest demonstration, summarised by a *tolerance radius*: the distance at which success is halfway between its level near a demonstration (the plateau) and its level far from every demonstration (the floor). As a reference, we measure the same radius for an *open-loop replay* (a demonstration's recorded actions executed with no policy or camera): a baseline for the same hardware and control stack. The policy, Action Chunking with Transformers (ACT) [1], is held fixed and is not the object of study. For the replay we find a well-defined radius, close to the median error of resets placed by eye. The policy succeeds at larger distances, but its success depends on location as well as distance, and its fitted radius is not established. Two layouts tested at 26 paired placements are not separated. We conclude with checklist entries.

Imitation policies are known to succeed near demonstrated placements and to fail a few centimetres away [2]–[5], so a reported success rate depends on where resets are placed and how accurately: the *reset distribution*. However, we are not aware of a paper that reports a measured accuracy for its object resets; a benchmark on the arm we use did not record positions, which "prevents analysis of spatial success patterns" [6].

Our earlier evaluation of the same task illustrates the problem (Fig. 1, Appendix D): of 31 trials whose starts we measured only afterwards, the 19 in cells with demonstrations had started within 5.6 mm of a trained demonstration and all succeeded, the 12 in held-out cells 31 to 47 mm away and one succeeded. No intermediate distance was tested: both rates were determined by where the cube was placed, and neither indicates the distance at which the policy begins to fail.

On a minimal task (one cube, one basket, a low-cost open-hardware arm [7]), each reset is placed by camera overlay within 2 mm of a target, at an adaptively chosen distance, and logged; a fitted success curve gives the radius. The second choice is often made by following a default: the LeRobot guide suggests "at least 50 episodes, with 10 episodes per location" [8]. We compare two layouts of 55 demonstrations: repeated starts at a few locations, or 55 distinct start positions. Both policies are tested at the same placements and compared with the success predicted from the fitted curve.

II. Related Work

Displacement. On real arms, policies have been tested at displacements no finer than about 25 mm [4], at shifts pooled into one rate [3], at placements 150 mm apart [2] or without stated distances [5], [9]. In simulation, DemoGen concludes that "the effective range can be approximated by the union

of the areas surrounding the demonstrated object placements" [2]; we test that hypothesis with a prediction from a fitted curve (Section III).

Evaluation. Absolute success rates "can easily be shifted up or down based on how difficult we make the task" [10]. Resets are made by hand on marks [4], by image overlay [10]–[12], on a grid board [13] or with a guiding tool [14].

Thresholds. Up-and-down staircases [15], [16], Bayesian adaptive placement [17], [18] and psychometric functions fitted with a lapse term [19] are standard in psychophysics [20], [21].

Layout. The documented default repeats locations [8], while a data-scaling study advises a position range "as extensive as possible" [22]. In simulation, 90 uniformly sampled demonstrations gave 66 % success over the workspace, against 29.5 % for 9 points with 10 each [23]. On the same arm, a study with a frozen vision encoder compared one, two and ten positions at 50 demonstrations [4].

Gap. We are not aware of a real-robot study that reports success against the measured distance to the nearest demonstration, estimates a displacement threshold for a manipulation policy with a staircase or a fitted psychometric function, or compares a clustered with a distinct-position layout at a fixed number of demonstrations. Like a recent study of controller gains [24], we measure such choices under controlled conditions.

III. Method

A. Rig and camera measurement

The rig is an SO-101 follower arm [7], front and wrist cameras in a light box, and LeRobot [25], [26] (settings in Appendix A). The task: pick up a 30 mm cube and place it in a basket. A grid of 5×3 cells of 50 mm is fixed to the table by two fiducial tags and is not drawn on it, so the policy sees no marks of position; no demonstration starts in its 4 *held-out* cells.

A *start position* is the cube's position as measured by the front camera, in millimetres in the table's frame. The same detector measures demonstrations and trials, each through a camera-to-table conversion solved from the two floor tags, so an offset common to both cancels in a distance; a printed millimetre scale checks the scale error that remains (Section IV-A). The independent variable is d: the distance from a trial's start position to the nearest one among the demonstrations its policy was trained on.

B. One budget, two layouts

Each dataset holds 55 demonstrations, 5 in each of the 11 other, *trained*, cells (Fig. 2). In the *clustered* layout the 5 demonstrations of a cell start at its centre. This is the structure that the LeRobot guide suggests [8], with fewer repeats per location. In the *spread* layout the demonstrations start at 55 distinct positions, at least 12 mm apart. One operator recorded both sets in four sessions on one day, following one procedure (Appendix A). We trained one ACT policy [1] on each set with identical settings and evaluated both at step 80,000, fixed in advance.

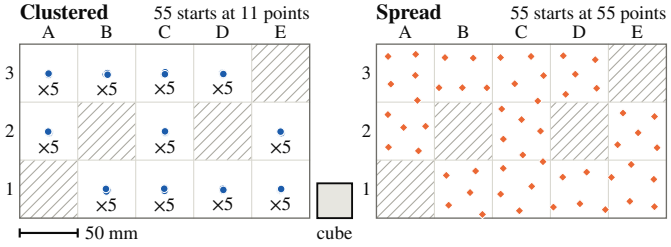


Fig. 2. The two layouts of one budget: measured start positions of the 55 demonstrations of the clustered set and of the spread set. Hatched cells are held out.

C. Trials

In a trial, as before each demonstration, an overlay guides the cube onto a planned target (Fig. 4); the placement is accepted within 2 mm and its measured position logged. The *condition* that then runs is the clustered policy, the spread policy, or a *replay*: a demonstration's recorded actions sent through the same control loop and safety gates, with no policy or camera. The operator scores each trial: only a cube released in the basket counts as a success. We replay 11 clustered demonstrations, one per trained cell, with the arm starting each replay from approximately that demonstration's initial pose; for a replay, d is measured from the start position of the replayed demonstration. For a requested distance L, the target is drawn uniformly among the admissible points whose distance to the nearest demonstration is L. One-up-one-down staircases [15], [16] choose L, larger after a success and smaller after a failure, so that trials concentrate where the outcome is uncertain. A condition's *threshold block* consists of two such staircases, one starting at a small distance and one at a large distance, and *anchor trials* at fixed distances: at $d = 0$ and, for the clustered policy, at 40 mm.

The clustered policy's block and the replay's ran interleaved in one shuffled order, over four consecutive sessions in about 91 minutes, so that both were exposed to the same state of the rig. The *workspace test* followed: 26 placements drawn uniformly, 13 over trained and 13 over held-out cells, each tested with both policies in immediate succession. The spread policy's own block was planned last and was not run. Each trial's condition was concealed from the operator by a code.

D. Analysis

The protocol, with its predictions and this analysis, was frozen before the first analysed trial (Appendix B). For each condition we fit

$$P(\text{success} \mid d) = \gamma + \frac{1 - \lambda - \gamma}{1 + \exp\big((d - r_{50})/w\big)}, \qquad (1)$$

with plateau $1 - \lambda$, the success level near a demonstration, where the lapse rate λ is the share of failures unrelated to distance [19]; floor γ, the probability of success far from every demonstration; width w of the transition; and tolerance radius

r_{50}, where success is halfway between plateau and floor. We report the posterior median of r_{50} with its 95 % posterior interval, or only a bound ("at most" or "at least") where the data do not bound the interval on one side. A *block fit* uses only a condition's own threshold block; all comparisons and predictions use block fits. A *pooled fit* also includes the policy's workspace-test trials and is reported as a check. As a *robustness check*, every fit is repeated with lapse rates up to 0.5 in place of 0.1; if the radius of either fit changes by more than 5 mm, we report that condition's radius as not established. We call one radius *larger* than another when the posterior probability that it exceeds the other is at least 0.95, and *smaller* when that probability is at most 0.05; otherwise the two are not distinguished. The *lookup prediction* for a policy at a placement is the clustered policy's fitted curve at the placement's distance to that policy's nearest demonstration.

IV. Results

A. The camera measurement and the resets

Against the printed scale, a measured 10 mm step deviated by 0.3 mm on average and by at most 0.8 mm. Measured distances were 1.019 times the true distances (1.012 to 1.025 across runs) and are reported uncorrected (Appendix A). Over the 108 scored trials, accepted placements were a median of 0.3 mm from their targets (maximum 0.6 mm), as read by the camera measurement that guided them. Placed by eye, using only a reference image and no overlay or readout, the cube was a median of 15.9 mm from its target (95th percentile 31.0 mm; illustrative, $n = 20$). Resets by image overlay [10]–[12] were not measured.

B. Replay and policy against distance

The replay's success falls within a narrow range of distance: it succeeded in 7 of 7 trials within 10 mm of the replayed demonstration's start, in 6 of 11 between 10 and 20 mm, and in 0 of 4 beyond (Table II, Fig. 3). Its tolerance radius is 16.5 mm (95 % interval 13.0 to 21.5), with a fitted width w of 2.0 mm; the robustness check changes the radius by 0.5 mm.

The clustered policy's success does not decrease steadily with distance: over all its scored trials it succeeded in 11 of 11 within 10 mm of a demonstration, in 6 of 10 between 10 and 20 mm, and in 9 of 12 between 30 and 40 mm, all of the last in held-out cells. The rise at 30 to 40 mm is within sampling variation, but the decline is gradual and location matters besides distance: inside the convex hull of its demonstrations the policy succeeded in 28 of 36 trials and outside it in 9 of 24 (a descriptive split fixed in advance), and in the two interior held-out cells in 13 of 18 trials against 3 of 12 in the two corner ones (an exploratory split). The block fit gives a radius of 39.0 mm (23.0 to 52.0) with a width of 13.7 mm; the pooled fit, which adds the policy's workspace-test trials, gives 36.0 mm (25.0 to 48.5). In the robustness check the block-fit radius changes by 3.5 mm, within our limit of 5 mm, but the pooled-fit radius changes by 6.0 mm, beyond it. By our pre-specified rule, which applies to either fit, the policy's radius is therefore *not established*, and every later result that uses the fitted curve is conditional on it (Appendix B, Table VI).

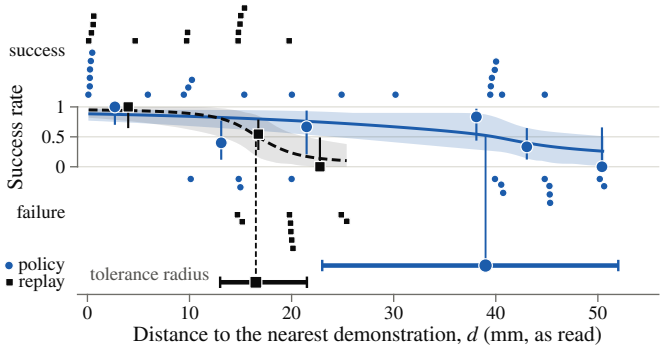


Fig. 3. Success against distance d for the clustered policy and the open-loop replay (threshold blocks; distances as measured, uncorrected). Small marks: single trials, in the row of their outcome. Large marks: success rates in 10 mm bins with Wilson intervals. Curves: posterior means of Eq. (1) with 95 % bands. Below: each r_{50} with its interval. Table II adds the workspace-test trials, so its counts differ.

TABLE I
Tolerance radius by condition, with the farthest success and the nearest failure of each threshold block (mm). First row of a condition: block fit; *pooled*: with the policy's workspace-test trials. The clustered policy's radius is not established (Table VI).

Condition	Scored (void)	Success at $d = 0$	Farthest success	Nearest failure	r_{50} (95 % interval)
Replay	22 (0)	4/4	19.8	14.7	16.5 (13.0–21.5)
Clustered	34 (0)	6/6	44.8	10.1	39.0 (23.0–52.0)
pooled	60 (0)				36.0 (25.0–48.5)
Spread, pooled	26 (0)				at least 19.5

The ordering of the two conditions is the same in each analysis. Under the pre-specified rule the policy's radius is the larger (posterior median of the difference 22.0 mm, 5.5 to 35.5; probability > 0.99). In the robustness check the two intervals still do not overlap (replay 13.0 to 23.5 mm, policy at least 26.0 mm). Without a model (Table II), within 20 mm the two are alike (policy 17 of 21 trials, replay 13 of 18), and beyond 20 mm the policy succeeded in 20 of 39 trials and the replay in 0 of 4. In each of these analyses the policy succeeds at larger displacements than open-loop execution of its own demonstrations on the same hardware, although its nearest failure (10.1 mm) is nearer than the replay's (14.7 mm).

Resets by eye. The median by-eye error, 15.9 mm, lies between 10 and 20 mm, where the clustered policy succeeded in 6 of 10 trials and the replay in 6 of 11, against 11 of 11 and 7 of 7 within 10 mm (Table II).

C. Two layouts at the same placements

In the workspace test the spread policy succeeded at 22 of 26 placements (Wilson interval [27] 66 to 94 %) and the clustered policy at 16 (43 to 78 %); the outcomes differed at 10 placements, 8 in favour of the spread policy and 2 of the clustered policy (exact McNemar test, two-sided, $p = 0.11$; Fig. 5 in Appendix C). The difference is in the predicted

TABLE II

SUCCESSES BY MEASURED DISTANCE d TO THE NEAREST DEMONSTRATION, OVER ALL SCORED TRIALS OF EACH CONDITION. DISTANCES WERE CHOSEN BY STAIRCASES OR FIXED IN ADVANCE, SO THESE ARE NOT RATES OVER A UNIFORM SAMPLE. A DASH: NO TRIAL. BEYOND 25 MM FROM A CLUSTERED DEMONSTRATION, A PLACEMENT LIES IN A CELL CORNER OR A HELD-OUT CELL.

		Clustered policy		Spread policy	
d (mm)	Replay	trained	held-out	trained	held-out
0–10	7/7	11/11	–	11/11	–
10–20	6/11	6/10	–	2/2	7/9
20–30	0/4	4/9	3/4	–	2/3
30–40	–	–	9/12	–	0/1
40–50	–	–	4/11	–	–
50–60	–	–	0/3	–	–

direction, but the test does not establish it: with 26 pairs it detects only large differences (Appendix B), and each layout was trained once, so this compares two trained policies and not two layouts in general. The *lookup prediction*, the clustered policy's block-fit curve applied to each placement's distance to a policy's nearest demonstration, expects 17.7 successes for the clustered policy (95 % predictive interval 10 to 24) and 21.0 for the spread policy (15 to 25). Both observed counts fall within these intervals, which are wide enough that only a large mismatch could have failed: this is weak evidence that distance alone explains the outcomes. Location matters for the spread policy too: inside the convex hull of its demonstrations it succeeded in 21 of 22 trials, and outside it in 1 of 4. The spread policy's threshold block was planned but omitted by the operator's decision, a deviation from the protocol (Appendix C); its workspace-test trials bound its radius only from below (at least 19.5 mm).

Of the four predictions stated before the data, two were met, one was met by its median alone and one was not met (Appendix B); the three deviations from the frozen protocol are listed in Appendix C.

V. PROPOSED CHECKLIST ENTRIES AND DISCUSSION

We propose five entries for the workshop's Design Choice Reporting Checklist, under *demonstrations* and *reset distributions*:

1) **Demonstration start states.** A map of where the object started in each demonstration, how the positions were set, and their scatter.
2) **Reset positions.** Each trial's measured object start and its distance to the nearest demonstration; the success counts by distance; the distances never tested.
3) **Reset accuracy.** The aid used to place the object, the measured placement error, and how the measurement was checked.
4) **Tolerance radius.** Where one curve describes the counts: the distance at which success is halfway between plateau and floor, with an interval or a one-sided bound and a check of the fit. In every case: the farthest success and the nearest failure.
5) **Open-loop reference.** The same for an open-loop replay of demonstrations, with its trials interleaved with the policy's.

The first three require a camera, two printed tags and a log; the last two require the trials of Section IV (34 and 22 trials here, a median of 1.3 minutes each).

Two relationships motivate these entries. First, reset accuracy relative to the distances at which success changes can alter a reported success rate: here, resets placed by eye had a median error of 15.9 mm ($n = 20$, one person), a distance at which the clustered policy succeeded in 6 of 10 trials, against 11 of 11 within 10 mm. Second, demonstration spacing relative to those distances affects how much of a workspace a dataset covers; here the test of the two layouts was inconclusive.

The procedure can be applied to another rig. Two further records concern defaults: the arm's software writes a position gain of 16 where, by its own source, the servo's default is 32 (Table III), and three data-pipeline faults of our earlier work pass the checks that were in place (Appendix E). The overlay, the planner, the trial logs, the measured start states and the frozen protocol are public at https://github.com/daniel-carrera/so101-tolerance-radius-study; the datasets are available on request.

VI. LIMITATIONS

This study has one rig, task, object and policy family, one budget of 55 demonstrations, one training run per layout and 108 trials: a difference between the two policies cannot be attributed to the layouts, and every number is specific to this rig and these runs: the earlier policy of Fig. 1 succeeded in 1 of 12 trials at 31 to 47 mm, the present one in 10 of 20 at the same distances. Beyond half a cell from the clustered demonstrations, distance is not separated from location (Table II). A code concealed each trial's condition, but the scorer also placed the cube and can often recognise a replay from its motion; rescoring 24 trials from video with identities concealed, the same person agreed in 24, a check of consistency and not an independent score. The cube's orientation was set by hand, unmeasured; the arm's start pose was logged, not controlled.

REFERENCES

[1] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware," in *Robotics: Science and Systems XIX*, 2023.

[2] Z. Xue, S. Deng, Z. Chen, Y. Wang, Z. Yuan, and H. Xu, "DemoGen: Synthetic Demonstration Generation for Data-Efficient Visuomotor Policy Learning," in *Robotics: Science and Systems XXI*, 2025.

[3] J. Zhao, W. Lu, D. Zhang, Y. Liu, Y. Liang, T. Zhang, Y. Cao, J. Xie, Y. Hu, S. Wang, J. Guo, D. Wang, and Y. Gao, "Do You Need Proprioceptive States in Visuomotor Policies?" *arXiv preprint arXiv:2509.18644*, 2025.

[4] Andresg324, "SO-101 × SmolVLA: How Demonstration-Collection Strategy Shapes Generalization," https://github.com/Andresg324/vla-data-strategy-study, 2026, GitHub repository, commit 9d98778 (2026-10-01). Accessed 2026-10-05.

[5] TamaraSumarac, "ACT vs Diffusion vs SmolVLA: How imitation learning policies generalize beyond their demonstrations," https://github.com/TamaraSumarac/act-diffusion-vla-so101, 2026, GitHub repository, commit c771d94 (2026-09-16). Accessed 2026-10-05.

[6] P. Selvaraj, L. Uttini, and V. Kuosmanen, "ArmnetBench v0.1: Parallel Real-World Evaluation of Manipulation Policies on a Low-Cost Arm Farm," *arXiv preprint arXiv:2607.24481*, 2026.

[7] TheRobotStudio, "Standard Open SO-100 & SO-101 Arms," https://github.com/TheRobotStudio/SO-ARM100, 2026, GitHub repository (created 2024-05-10), commit 5f6d2b8 (2026-09-23). Accessed 2026-10-05.

[8] Hugging Face, "Imitation Learning on Real-World Robots," https://huggingface.co/docs/lerobot/il_robots, 2026, LeRobot documentation, section "Tips for gathering data"; source file docs/source/il_robots.mdx, line 407, at commit 6adf515 (2026-08-13) and on the main branch. Accessed 2026-10-05.

[9] J. Long, W. Cai, and W. Zhi, "Register-Routed Delayed Fusion: Rewiring Shortcut-Prone Observation Fusion in Visuomotor Imitation," *arXiv preprint arXiv:2610.02813*, 2026.

[10] TRI LBM Team *et al.*, "A Careful Examination of Large Behavior Models for Multitask Dexterous Manipulation," *arXiv preprint arXiv:2507.05331*, 2025.

[11] H. Kress-Gazit, K. Hashimoto, N. Kuppuswamy, P. Shah, P. Horgan, G. Richardson, S. Feng, and B. Burchfiel, "Robot Learning as an Empirical Science: Best Practices for Policy Evaluation," *arXiv preprint arXiv:2409.09491*, 2024.

[12] A. S. Huang, J. Zhang, S. Tang, and Y. Xiang, "VLA-REPLICA: A Low-Cost, Reproducible Benchmark for Real-World Evaluation of Vision-Language-Action Models," *arXiv preprint arXiv:2605.20774*, 2026.

[13] S. Jin, Y. Wang, Y. Duan, D. Wu, G. Dong, X. Liu, X. Li, H. Jia, Z. Zhang, T. Wang, Z. Jia, Y. Yao, C. Bai, Zhaxizhuoma, S. Liu, N. Cao, J. Wang, C. Yu, and Y. Ding, "UMI-Bench 1.0: An Open and Reproducible Real-World Benchmark for Tabletop Robotic Manipulation with UMI Data," *arXiv preprint arXiv:2606.10382*, 2026.

[14] M. Heo, Y. Lee, D. Lee, and J. J. Lim, "FurnitureBench: Reproducible Real-World Benchmark for Long-Horizon Complex Manipulation," in *Robotics: Science and Systems XIX*, 2023.

[15] W. J. Dixon and A. M. Mood, "A Method for Obtaining and Analyzing Sensitivity Data," *Journal of the American Statistical Association*, vol. 43, no. 241, pp. 109–126, 1948.

[16] H. Levitt, "Transformed Up-Down Methods in Psychoacoustics," *The Journal of the Acoustical Society of America*, vol. 49, no. 2B, pp. 467–477, 1971.

[17] A. B. Watson and D. G. Pelli, "QUEST: A Bayesian adaptive psychometric method," *Perception & Psychophysics*, vol. 33, no. 2, pp. 113–120, 1983.

[18] L. L. Kontsevich and C. W. Tyler, "Bayesian adaptive estimation of psychometric slope and threshold," *Vision Research*, vol. 39, no. 16, pp. 2729–2737, 1999.

[19] F. A. Wichmann and N. J. Hill, "The psychometric function: I. Fitting, sampling, and goodness of fit," *Perception & Psychophysics*, vol. 63, no. 8, pp. 1293–1313, 2001.

[20] B. Treutwein, "Adaptive psychophysical procedures," *Vision Research*, vol. 35, no. 17, pp. 2503–2522, 1995.

[21] M. R. Leek, "Adaptive procedures in psychophysical research," *Perception & Psychophysics*, vol. 63, no. 8, pp. 1279–1292, 2001.

[22] F. Lin, Y. Hu, P. Sheng, C. Wen, J. You, and Y. Gao, "Data Scaling Laws in Imitation Learning for Robotic Manipulation," *arXiv preprint arXiv:2410.18647*, 2024.

[23] H. Wang, C. B. Chen, Y. Yue, D. Tao, T. Guo, S. Xie, D. Huang, S. Song, G. Yao, and G. Huang, "MOVE: A Simple Motion-Based Data Collection Paradigm for Spatial Generalization in Robotic Manipulation," *arXiv preprint arXiv:2512.04813*, 2025.

[24] A. Bronars, Y. Park, and P. Agrawal, "Tune to Learn: How Controller Gains Shape Robot Policy Learning," *arXiv preprint arXiv:2604.02523*, 2026.

[25] R. Cadene, S. Alibert, A. Soare, Q. Gallouedec, A. Zouitine, S. Palma, P. Kooijmans, M. Aractingi, M. Shukor, D. Aubakirova, M. Russi, F. Capuano, C. Pascal, J. Choghari, K. Meftah, M. Ellerbach, J. Moss, and T. Wolf, "LeRobot: State-of-the-art Machine Learning for Real-World Robotics in Pytorch," https://github.com/huggingface/lerobot, 2024, GitHub repository; citation block of README.md at commit 6adf515 (2026-08-13). Accessed 2026-10-05.

[26] R. Cadene, S. Alibert, F. Capuano, M. Aractingi, A. Zouitine, P. Kooijmans, J. Choghari, M. Russi, C. Pascal, S. Palma, M. Shukor, J. Moss, A. Soare, D. Aubakirova, Q. Lhoest, Q. Gallouédec, and T. Wolf, "LeRobot: An Open-Source Library for End-to-End Robot Learning," *arXiv preprint arXiv:2602.22818*, 2026, the LeRobot repository's README gives the venue as ICLR 2026.

[27] E. B. Wilson, "Probable Inference, the Law of Succession, and Statistical Inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.

Appendix A
The rig's settings

The full protocol, the trial log, the code and a map of every trial are at https://github.com/daniel-carrera/so101-tolerance-radius-study. Tables III and IV list the rig's settings in the categories of the workshop's checklist; the repository's `paper/numbers_by_hand.tex` gives each in full with its source. None of these settings was varied, so this study cannot determine how the tolerance radius depends on them.

Measurement. The grid is not drawn on the floor; it appears only in the overlay (Fig. 4). Two AprilTags outside the grid, located once from a printed sheet of six tags, tie it to the floor, and each session solves the conversion to table millimetres again from them. The detector reports the centre of the cube's visible outline, not of its footprint, so distances carry a scale factor a few percent above one. Over its 4 runs the scale check gave a factor between 1.012 and 1.025 (mean 1.019). The rule fixed in advance: if the runs agree within one percentage point, every distance is divided by the mean factor; otherwise distances are left as measured and the range of the factor is reported as the instrument's accuracy. For repeatability, the cube was pushed 11 times against a fixed stop; the readings stayed within 0.3 mm of their mean.

Recording. One operator recorded both sets on one day with the same task text, time limit, cube orientation, start pose, approach and grasp, in rounds of one episode per trained cell in shuffled order. Before training, each dataset passed a start-state audit: every episode in its planned cell and within 3 mm of its target.

Appendix B
The protocol: predictions, scoring and analysis

The protocol (version 1.0) was frozen before the first counted trial, and only counted trials are analysed; pilot trials, run before the freeze to test the procedure, were logged apart. The predictions below were written three days before the first counted trial and were not changed.

A. Predictions stated before the data

Every radius here is a block fit.

1) *Curve.* The clustered policy succeeds at least 5 times in 6 at $d = 0$, at most once in 4 at 40 mm, and has a radius of 5 to 40 mm. *Observed:* 6 of 6, 1 of 4, and 39.0 mm (23.0 to 52.0): met by the median alone.
2) *Hardware.* The policy's radius exceeds the replay's. *Observed:* a difference of 22.0 mm (5.5 to 35.5), probability > 0.99: met, though the policy's radius is not established.
3) *Layout.* In the workspace test, the spread policy succeeds more often. *Observed:* 22 against 16 of 26 ($p = 0.11$): not met.

TABLE III
THE RIG'S SETTINGS, IN THE CATEGORIES OF THE WORKSHOP'S DESIGN CHOICE REPORTING CHECKLIST (CONTINUED IN TABLE IV).

Category	Setting	Value
Hardware and embodiment	Arm	SO-101 follower and leader [7], both built by the operator from one seller's kit (WowRobo); the follower's base is clamped at the front of the light box
	Gripper	The kit's standard printed jaws, with nothing added to them; fully open, the jaw tips are 151 mm apart (the operator's ruler)
	Object and basket	Cube of 30 mm: painted wood, orange, one face with rounded edges (placed face down); mass 20 g. Basket: grey plastic, 256×201 mm at the rim and 104 mm tall, taped down beyond the grid's column A
	Workspace	Grid of 5×3 cells of 50 mm, fixed to the table by two tags (AprilTag 36h11, ids 0 and 1, printed at a nominal 60 mm (measured: 59.7×60.0 and 59.8×60.0 mm)); an 81 cm cubic photography light box with three LED panels under the top, behind a white diffuser
Motors and actuation	Servos	Feetech STS3215; follower: the 12 V variant, geared 1/345, on a 12 V, 8 A supply; leader: the kit's 7.4 V variants
Kinematics	Joints, calibration	Five arm joints and a gripper; calibrated with LeRobot's procedure, files written on 2026-07-03 and unchanged since
Controller gains and rates	Servo position loop	P = 16, I = 0, D = 32 on all six servos, written by LeRobot at every connection; by LeRobot's source the servo's own default P is 32
	Control rate	30 Hz requested; 29.1 Hz achieved in policy trials
	Limits on a command	Per-step clamp of 15 degrees; a policy's first command within 8 degrees of the arm's pose on every joint, a replay's within 5; stall rule: an attempt ends when one joint is at least 13.5 degrees from its command (0.9 of the clamp) and no joint has moved 1.5 degrees over the last 30 steps
Action spaces	Action	A position target for each of the six servos at every control step; a demonstration's actions are the leader arm's joint positions
Observations	Cameras	Front: Logitech HD Pro Webcam C920, inverted on the box's top rear pipe, lens 674 mm above the floor and 810 mm from the middle of the grid; settings fixed by hand. Wrist: the kit's 2-megapixel USB camera on its printed mount. Both 640×480 pixels at 30 Hz
	Other inputs	The follower's six joint positions, in the action's units, at the current step only

4) *Lookup.* Each policy's workspace-test count lies in the lookup prediction's 95 % interval. *Observed:* 16 in 10 to 24 (clustered), 22 in 15 to 25 (spread): met, with wide intervals.

A fifth item, the spread layout's own radius, was to be reported as an estimate; its block was not run (Appendix C).

B. The blocks

Table V gives each block. Staircases move in steps of 10 mm until their first reversal and 5 mm after it, and a requested distance is met within 0.5 mm. The held-out cells are A1, B2, D2 and E3. The replay block plays back 11 demonstrations of the clustered set, one from each trained cell, chosen by the planner's seed before any trial, and replays each twice. In the replay pilot, each of the 11 demonstrations was replayed once from its own start position: ten succeeded, and the eleventh attempt was ended by an accidental key press and is logged as a failure. In the workspace test the two trials of a pair ran in immediate succession on the same target.

C. Scoring

The operator scored each attempt at the rig; success requires the cube released in the basket within the time limit. A trial is void when it could not test its condition: the placement was later found outside tolerance, a camera or motor fault ended the attempt, a person interfered, the safety gate refused the start three times, or a key press ended the attempt within one second of its start. A void trial remains in the log with its reason, and its place in the plan is repeated with the same target under a new number and code. A stop caused by the condition's own motion after the handover (the per-step clamp, the stall rule) is a failure. After the last trial, one scored trial in five of each condition of each block (24 of 108) was drawn at random and scored again from its video alone, with no condition, distance or first score shown. No second person was available, so the operator gave both scores; they agree in 24 of 24 trials. The score given at the rig remains the outcome.

D. Analysis fixed in advance

- *Estimate.* Eq. (1), with a posterior on a grid: a flat prior on r_{50} from 0 to 80 mm in steps of 0.5 mm, a flat prior on $\log w$ over 30 values from 0.5 to 20 mm, and $\lambda \in \{0, 0.02, 0.05, 0.10\}$ and $\gamma \in \{0, 0.05, 0.10\}$, equally weighted. We report the posterior median of r_{50} and its equal-tailed 95 % interval; the staircases only choose where to test.
- *Open intervals.* When the posterior at an end of the grid is at least half its value at the interval's bound on that side, we report "at most" or "at least" that bound, and not a radius.
- *Plateau check.* Every fit is made again with $\lambda \in \{0, 0.02, 0.05, 0.10, 0.20, 0.35, 0.50\}$ (Table VI). If the two medians differ by more than 5 mm, the condition's radius is not established, and every result that rests on it is still computed by its rule and reported with that caveat.
- *Comparisons.* For two radii, the posterior probability that one exceeds the other; "larger" means at least 0.95. For the workspace test, the exact McNemar test at a two-sided p below 0.05. With 26 pairs and true success rates of 40 and 90 %, that test reaches $p < 0.05$ in 96 % of cases

TABLE IV
THE RIG'S SETTINGS, CONTINUED: DEMONSTRATIONS, THE POLICY, AND THE RESET DISTRIBUTIONS.

Category	Setting	Value
Demonstrations	Interface	Leader arm, recorded at 30 Hz (median 29.99 achieved); episodes of 17.3 to 26.0 s, ended by the operator's key
	Budget and layout	55 per dataset. Clustered: 5 at each of 11 points. Spread: 55 positions, planned at least 12 mm apart (measured: at least 17.5 mm)
	Recording	4 recording sessions on one day, in the order clustered, spread, spread, clustered; 2 clustered and 1 spread episodes were recorded again
	Start states	Set with the overlay, then measured from each episode's first frame: 0.7 mm from the planned target at the median, 2.7 mm at most
	Trained on	55 clustered and 55 spread episodes; held out for an evaluation loss: none
Policy (held fixed; not a checklist category)	Model and training	ACT [1], 51.6 M parameters, ImageNet-pretrained ResNet-18 backbone, chunk of 100 steps, batch 32, learning rate 10^{-5}, seed 1000, image augmentation off; one run per dataset on one A100 GPU, LeRobot 0.6.1; evaluated at step 80,000
	Execution	Temporal ensembling with coefficient 0.01 and inference at every step; attempts limited to 45 s; LeRobot at commit 6adf51511b
Simulation		None: every trial ran on the arm
Reward design		None: imitation learning. Outcomes were scored by a person on a graded scale, with success only for a cube released in the basket
Reset distributions	Object	Placed with the overlay within 2 mm of a planned target; accepted placements 0.3 mm from the target at the median, 0.6 mm at most; every measured position and its d logged
	Arm	Set by the operator with the leader arm before every trial, folded with the gripper closed; no pose is stored or commanded
	Camera	Each session solves the camera-to-table conversion from the two floor tags; no attempt is made once the image has moved more than 1 mm from that solve (the reading before an attempt: 0.57 mm at the median, 0.99 mm at most)

TABLE V
THE BLOCKS AS PLANNED. DISTANCES IN MILLIMETRES.

Block	Trials at $d = 0$	Trials at 40 mm	Two staircases: starts; trials each	Largest distance
Replay	4	–	5 and 25; 9	50
Clustered policy	6	4	10 and 40; 12	50
Spread policy	4	–	5 and 25; 10	50
Workspace test	26 placements, each with both policies: 13 in trained cells, 13 in held-out cells			

TABLE VI
THE PLATEAU CHECK: EACH FIT WITH LAPSE VALUES UP TO 0.10, AND AGAIN UP TO 0.50. POSTERIOR MEDIAN AND 95 % INTERVAL OF r_{50} IN MM; AN INTERVAL NOT BOUNDED ABOVE IS GIVEN AS ITS LOWER BOUND. SHIFT: SECOND MEDIAN MINUS FIRST. SECOND COLUMN: SUCCESSES AT $d = 0$.

Fit	At $d = 0$	$\lambda \leq 0.10$ median	$\lambda \leq 0.10$ interval	$\lambda \leq 0.50$ median	$\lambda \leq 0.50$ interval	Shift
Replay, block	4/4	16.5	13.0–21.5	17.0	13.0–23.5	0.5
Clustered, block	6/6	39.0	23.0–52.0	42.5	≥ 26.0	3.5
Clustered, pooled	6/6	36.0	25.0–48.5	42.0	27.5–58.0	6.0
Spread, pooled	none	28.5	≥ 19.5	29.0	≥ 20.0	0.5

(computed before any trial), and it cannot reach it with fewer than six discordant pairs, so it detects only large differences.

- *Without a model, and exclusions.* Success counts in 10 mm bins of d with Wilson intervals [27]. No trial is excluded except void and pilot trials, and no outcome is changed after the fact.

TABLE VII
TRIALS PLANNED, SCORED AND VOID, BY BLOCK.

Block	Planned	Scored	Void
Replay	22	22	0
Clustered policy, threshold	34	34	0
Workspace test, both policies	52	52	0
Spread policy, threshold	24	0	0
All	132	108	0

APPENDIX C
DEVIATIONS, TRIAL COUNTS AND VOID TRIALS

The decision log records 3 deviations from the frozen protocol. First, the spread policy's threshold block (24 trials) was not run: the operator gave the remaining time to the by-eye resets, the repeatability check and the second scoring, before any count by condition had been computed. The spread policy therefore has no block fit. Second, the repeatability check, which the protocol places before any trial, was run after the last trial on the same day. Third, the camera check planned at the end of each block was made instead at the start of the next session, and both passed (0.8 and 0.7 mm from the stored calibration; the limit is 5 mm). Table VII gives the trials planned, scored and void in each block. Void trials: none. Every failure carries the label "grasp". Number of days on which trials ran: 1. Number of people who placed the cube and scored: 1. Fig. 5 shows the workspace test; both policies succeeded at 14 placements and neither at 2.

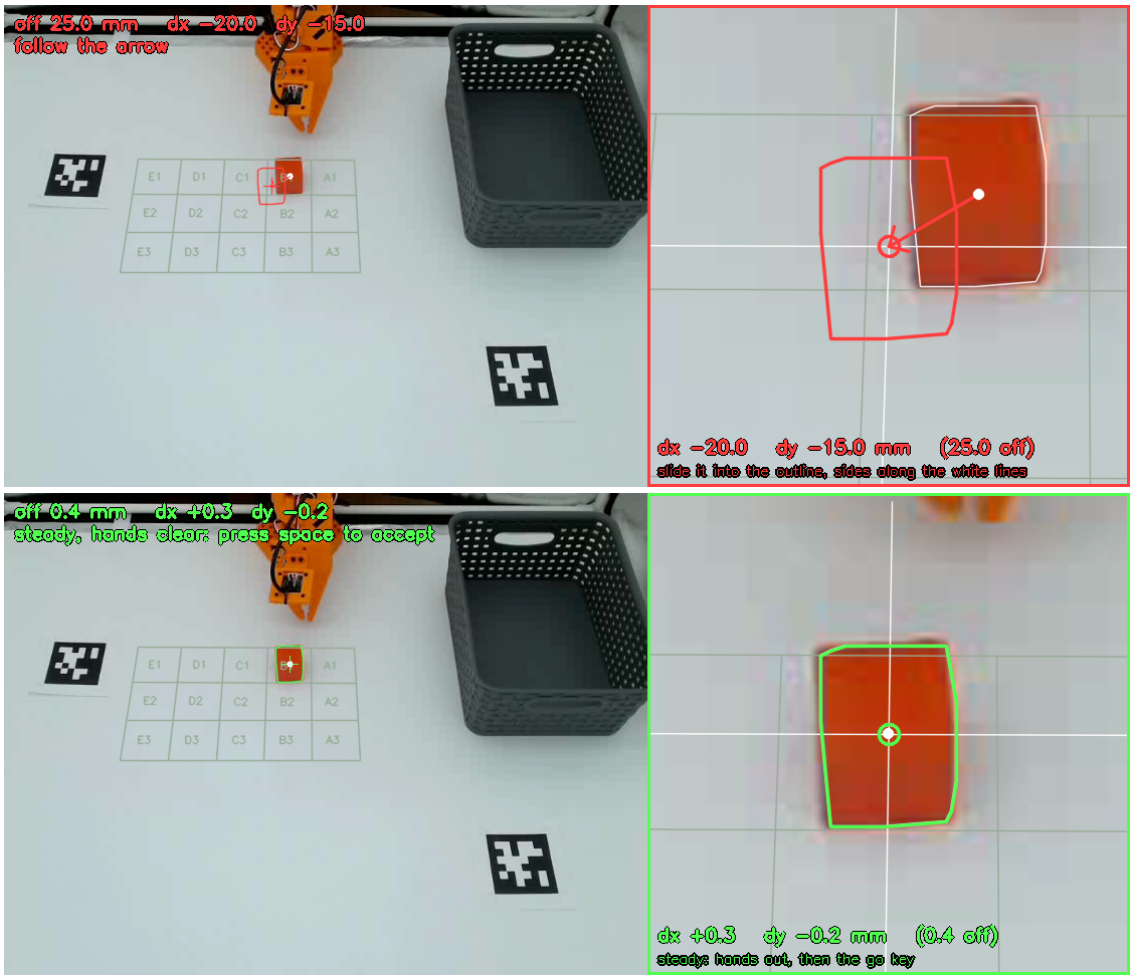


Fig. 4. The placement overlay, rendered from the first frame of a counted trial's video. Left: the front camera's image with the grid, the two floor tags, the arm and the basket. Right: the target, magnified. Top: the cube 25 mm from an illustrative target. Bottom: within tolerance of the trial's own target (0.4 mm).

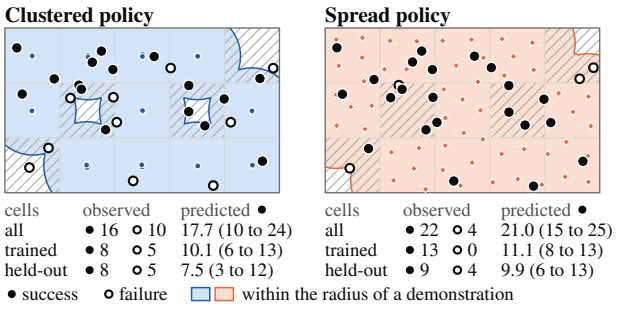


Fig. 5. The workspace test: each policy's demonstrations and its outcome at each of the 26 shared placements. The discs have the clustered policy's fitted r_{50}, which is not established. Below: observed successes, and those the lookup prediction expects.

Appendix D
The earlier evaluation, re-analysed

The earlier evaluation of Fig. 1 tested one ACT policy on the same task and arm, with the front camera on a tripod instead of in the light box. Its trials did not log the cube's start; we measured it afterwards from the trials' videos, with distances to the nearest of the 49 demonstrations the policy was trained on. The 19 trials in cells with demonstrations started 0.6 to 5.6 mm from the nearest and all succeeded (Wilson 95 % interval 83 to 100 %); the 12 in held-out cells started 31 to 47 mm away and one succeeded, at 40 mm (1.5 to 35 %). That evaluation cannot separate distance from cell, since every near trial was in a cell with demonstrations and every far trial in a held-out one; its 31 trials are 12 scenes run two or three times each, so the intervals above are too narrow; and its distances carry a scale error of a few percent that was not measured.

Appendix E
Faults in the data pipeline of the earlier work

The earlier project's own records show three faults, each of which passes a check that the project had and each of which the project itself found within a day (Table VIII); we did not measure their effect on the policy. In this study's two datasets, a scan written for this rig flagged 9 frames in 6 of the 110 episodes; 7 are torn, all in the top rows of the front camera's

image, outside the area of the task, and all episodes were used for training.

TABLE VIII
FAULTS RECORDED IN THE EARLIER PROJECT'S DATA AND TRIALS.

Fault	Extent	Checks that pass despite it
Torn video frames: one frame composed of parts of two frames	31 frames flagged in 18 of the 55 episodes behind the earlier evaluation	The dataset check, which decodes the first and last frame of every episode, and ffmpeg's error report: both pass on these files
Episodes captured more slowly than their time stamps say	16 episodes at about 28 Hz, stamped 30 Hz	No warning at recording, where the shortfall was logged at debug level only; the dataset check of frame pacing, which reads nominal time stamps and still reports that the loop never stalled
An execution setting off without notice: temporal ensembling of the policy's actions	2 trials, kept apart from the reported ones	A unit test of the option's default, which read the value written in the source and not what a run parsed